

DIABETES MELLITUS — Clinical Revision Table

Bold = high-yield. Glucose values in mg/dL. Covers definition, types, diagnosis, drugs, insulin, emergencies-prep.

1 — OVERVIEW & HOMEOSTASIS

DM = vascular disease	Lack / reduced effectiveness of insulin. Microvascular : retinopathy, nephropathy, neuropathy. Macrovascular : stroke, MI, renovascular, limb ischaemia. 415M (2015) → 642M by 2040.
Glucose control	Post-meal : ↑insulin → suppress hepatic glucose output, ↑muscle uptake, glycogen synthesis, ↑lipogenesis, suppress lipolysis. Fasting : ↓insulin → gluconeogenesis (glycogen, glycerol, amino acids). Brain depends on liver glucose.

2 — TYPE 1 vs TYPE 2

	TYPE 1	TYPE 2
Onset	Adolescent (any age)	Usually >40 (now teens too)
Cause	Autoimmune β-cell destruction → insulin deficiency	Insulin resistance + ↓secretion ; obesity, inactivity, alcohol/calorie excess
Antibodies	Insulin, GAD , IA-2, ZnT8 (anti-GAD+IA-2 ~85%)	None
Genetics	HLA DR3±DR4 ; twin concordance ~30%	No HLA; ≥80% twin concordance (stronger genetic)
Body / risk	Ketoacidosis + weight loss ; must have insulin	Obesity, metabolic syndrome
Presentation	Polyuria, polydipsia, ↓weight, ketosis	Asymptomatic or complication (e.g. MI)
Variants	LADA = slow type 1 in adults (measure islet Ab if older+ketotic+slim)	MODY = autosomal dominant young; C-peptide : low=T1, high=insulin resistance

3 — PREDIABETES, OTHER CAUSES, METABOLIC SYNDROME

IGT	Fasting <126 + OGTT 2h 140–199 (post-prandial defect)
IFG	Fasting 110–125 (fasting defect; do OGTT). Lower progression risk than IGT. IFG + HbA1c 5.5–6.4% → ~25% develop DM
Secondary causes	Drugs : steroids, anti-HIV, antipsychotics. Pancreatic : pancreatitis, surgery, trauma, haemochromatosis, CF, cancer. Hormone excess : acromegaly, Cushing's, glucagonoma, pheochromocytoma, thyrotoxicosis. Gestational
Metabolic syndrome	≥3 of 5 : central obesity (BMI>30 / waist >40"M, >35"F); BP ≥130/85; TG ≥150; HDL <40M/<50F; fasting glucose ≥100. Tx: exercise, weight loss, treat components

4 — DIAGNOSIS (WHO)

Criteria	Symptoms + 1 raised glucose (fasting ≥126 OR random ≥200) OR raised on 2 occasions (no symptoms) OR OGTT 2h ≥200. Symptoms: polyuria, polydipsia, weight loss, blurring, genital thrush, lethargy
HbA1c	<5.7% normal · 5.7–6.4% prediabetes · ≥6.5% diabetes . = avg glucose 2–3 mo
HbA1c caveats	Avoid: pregnancy, haemoglobinopathies . Falsely LOW: cirrhosis, recent blood loss, transfusion → use fructosamine
OGTT method	Unrestricted carbs 3d → fast ≥8h → rest → seated/no smoking → glucose before + 2h after 75g glucose

5 — MANAGEMENT GENERAL & TARGETS

General	Education + lifestyle (exercise ↑insulin sensitivity), ↓sat fat/sugar, ↑starch. Statin (atorva/rosuva) + BP control + foot care. Bariatric = cure in selected. Type 1 → urgent insulin + referral ; Type 2 → lifestyle first, add orals
Glycaemic targets	HbA1c <7% · preprandial 80–130 · peak postprandial <180. Individualize (age, comorbidity, hypo-unawareness)

6 — ORAL AGENTS

Class	Mechanism	Key points / SE
Metformin (biguanide)	Insulin sensitizer — ↓hepatic glucose output	No hypo, weight-friendly. FIRST-LINE all . SE: GI. Avoid eGFR ≤36 (lactic acidosis)
Sulfonylureas	Insulin secretagogue (gliclazide, glimepiride, glipizide)	SE: hypoglycaemia, weight gain . Avoid glibenclamide in elderly
DPP-4 i / gliptins	Block DPP-4 → ↑incretin (GLP-1, GIP) → ↑insulin	Well tolerated, weight neutral (sita/vilda/saxa/lina/alogliptin)
Glitazones (TZD)	↑insulin sensitivity	SE: fractures, fluid retention , ↑LFT. CI: heart failure, osteoporosis
SGLT2 i (-gliflozin)	Block renal glucose reabsorption → glycosuria	Empagliflozin ↓CV mortality . SE: genital fungal, euglycaemic DKA

7 — INSULIN & GLP-1 ANALOGUES

Insulin types	Rapid: lispro, aspart, glulisine. Long-acting: glargine, detemir, degludec. Isophane (peak 4–12h). Pre-mixed (NovoMix 30). Strength 100 U/mL
Regimens	BD biphasic (premixed, T2/regular T1) · QDS basal-bolus (T1 flexible) · once-daily basal (switching from tablets, retain metformin) · DAFNE = flexible dosing
Sick-day rules	NEVER stop insulin. Illness ↑requirements. Maintain calories (milk). Check glucose ≥4×/day + ketonuria. ↑dose if rising. Admit if vomiting/dehydrated/ketotic/child/pregnant. Pump if disabling hypos
GLP-1 analogues	Incretin mimetic: ↑insulin, ↓glucagon, slow gastric emptying. BMI >35. Continue only if HbA1c ↓≥1% + weight ↓≥3% in 6mo. Daily (liraglutide) / weekly (semaglutide, dulaglutide); oral semaglutide. SE: nausea, small pancreatitis risk